Strategies for Enhancing Courtyard Thermal Comfort in Hot Arid Phoenix: The Secret Garden on ASU Campus

Muge Unal[1], Ariane Middel[2]

[1]Fırat University, Landscape Architecture Department, Elazig/Türkiye · mugeunal@firat.edu.tr
[2]Arizona State University, School of Arts, Media and Engineering (AME), Tempe, Arizona/USA

Abstract: Courtyards possess unique geometric features that regulate sun exposure, wind, and humidity in hot climates. This study examines how surface materials and shading affect thermal comfort in courtyards in the hot arid climate of Phoenix, Arizona, through human-biometeorological observations and microclimate modeling. The Herberger Institute for Design and the Arts Hall (Secret Garden) on Arizona State University's main campus were selected as case studies. The thermal environment of these courtyards was first evaluated for a typical hot and dry summer day in June 2023 using the ENVI-met 5.0.1 microclimate model. On-site measurements of key microclimatic variables, including air temperature (Ta), relative humidity (RH), wind speed (WS), and mean radiant temperature (MRT), were collected using the MaRTy platform to validate the model. Two scenario groups were then developed: the first tested various surface materials (concrete, grass, and water), and the second explored the impact of 3D vegetation configurations (central and bordered planting). Statistical analyses, including ANOVA and Tukey's HSD tests, were conducted to compare the spatial differences in thermal exposure and comfort across the scenarios.

Keywords: Courtyard design, microclimate, MaRTy, ENVI-met, heat mitigation strategies

1 Introduction

In recent years, meteorological studies have increasingly focused on micro-scale evaluations, highlighting the critical influence of urban design on thermal conditions (GELETIČ et al. 2019, HU et al. 2019). Courtyards, historically used to enhance thermal comfort, are valued for modulating airflow, providing shade, and reducing heat retention. Their design has traditionally reflected local climatic and cultural needs, and today, they remain integral to residential, educational, and healthcare architecture for their passive cooling benefits (GHAFFARIANHOSEINI et al. 2015).

Research on courtyards often explores their morphology, geometry, surface materials, and shading effects, employing tools like ENVI-met, Ray-Man, and EnergyPlus to evaluate thermal performance in diverse climates (GHAFFARIANHOSEINI et al. 2015, CRANK et al.2020). CFD models, calibrated with field data, are particularly effective in simulating urban microclimates, including wind and temperature variations (MIDDEL et al. 2014, TALEGHANI et al. 2014). ENVI-met is a high-resolution non-hydrostatic microclimate model developed by Michael Bruse, which is widely used for analyzing microclimates and thermal comfort indices. Recent studies have employed ENVI-met to assess how urban physical characteristics impact thermal comfort, particularly in hot-arid and hot-humid areas (UNAL CILEK & CILEK 2021, UNAL & USLU 2022). The model is frequently utilized to examine how courtyard attributes like orientation, aspect ratio, SVF, building height, shade, surface cover, and vegetation influence outdoor thermal comfort (AKBARI et al. 2021, BERKOVIC et al. 2012, LIU et al. 2024, MARCHI et al. 2023, SÖZEN & ORAL 2019, TALEGHANI et al. 2014). Studies also highlight the

Journal of Digital Landscape Architecture, 10-2025, pp. 173-180. © Wichmann Verlag, VDE VERLAG GMBH · Berlin · Offenbach. ISBN 978-3-87907-754-0, ISSN 2367-4253, e-ISSN 2511-624X, doi:10.14627/537754016.
This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution license (http://creativecommons.org/licenses/by-nd/4.0/).

benefits of material selection, lawns (MUSHTAHA et al. 2021, SALUR 2016, SOFLAEI et al. 2017), and trees or shading elements on thermal conditions (AKBARI et al. 2021, DIZ-MELLADO et al. 2021, SOFLAEI et al. 2020). However, model validation remains limited, with RMSE often exceeding ±5 °C (WU et al. 2023). Validation with in-situ data is essential for accurate interpretation (CRANK et al. 2020).

This study's originality lies in the detailed examination of the thermal performance of courtyards in hot arid climate regions such as Phoenix, Arizona. The literature on micro-climate dynamics and design interventions specific for Arizona remains limited. Furthermore, our integrated methodological approach, supported by both advanced modelling and field measurements, provides a significant methodological contribution to existing re-search. This study aims to evaluate the microclimate performance of selected courtyards using the ENVI-met 5.0.1 model and MaRTy platform by analyzing key microclimate variables, including air temperature, relative humidity, wind speed, and mean radiant temperature. Additionally, it investigates the impact of different surface materials (concrete, grass, and water) and 3D vegetation configurations (central and bordered planting) on thermal comfort. The strength of our methodology, particularly in the use of scenario-based analysis, ANOVA, and Tukey's HSD tests. These approaches allow for a robust statistical comparison of thermal exposure and comfort variations across different design scenarios. It aims to provide guidelines for enhancing outdoor thermal comfort.

2 Materials and Methods

2.1 Study Area

This study was conducted on the Arizona State University (ASU) main campus in Tempe, Arizona, located in Maricopa County (33°26'54″N, 112°04'26″W). Covering 2.7 km² near downtown Tempe, the area features a tropical and subtropical desert climate (BWh) per the Köppen-Geiger classification, with low annual rainfall (203.2 mm), hot summers (41-53°C in July-August), and mild winters (6–13°C in January–February) (AZMET 2023, CILEK et al. 2024). Vegetation includes drought-resistant species and irrigated lawns. The ASU campus was selected for its accessible measurement facilities and similarly designed courtyards. The courtyard at the Herberger Institute for Design and the Arts Hall (Secret Garden) is north-south oriented and enclosed by a 54 × 64 m² building with an 8 m height. It contains trees, shrubs, cacti, succulents, and grass, with a courtyard width of 36 m, length of 62 m, and a low height-to-width (H/W) ratio of 0.22 for both East-West and South-North orientations. Sky View Factor (SVF) ranges from 0.09-0.18 in vegetated areas and 0.54-0.63 at courtyard centers (Fig. 1).

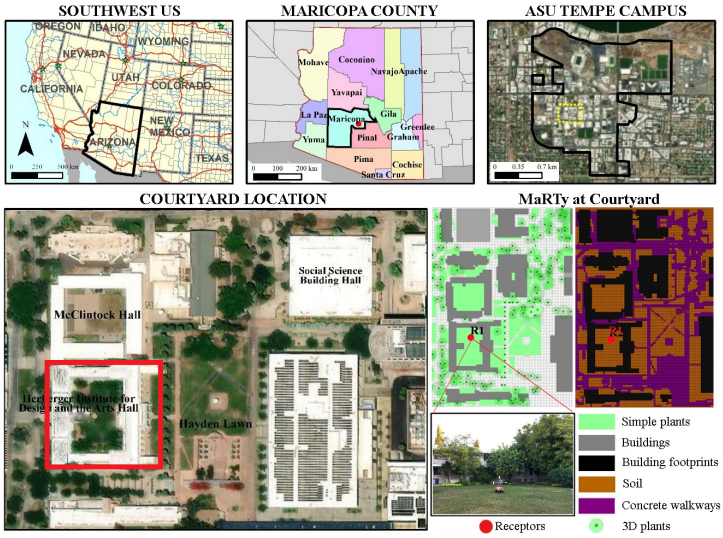


Fig. 1: Herberger Institute for Design and the Arts Hall

2.2 Methods

Table 1: Matrix of 18 scenarios combining of courtyard with changes in ground cover, vegetation configuration, and plant placement strategies

	100% grass	85% grass 15% water	100% concrete	85% concrete 15% water	15% grass 85% concrete	45% concrete 40% grass 15% water
No 3D plants						
3D plants centered						
3D plants border						

The study employed a five-step methodology. First, micrometeorological stations (MaRTy human-biometeorological instrument platforms, MIDDEL & KRAYENHOFF 2019) were installed in and outside courtyards to monitor air temperature (Ta), wind speed (WS), wind direction (WD), relative humidity (RH), and mean radiant temperature (MRT). These hourly observations, recorded at 2-second intervals and averaged by location, were used to initialize and evaluate ENVI-met simulations modeling a 158 m x 97 m area with a 2 x 2 m grid. Data

were collected on June 20, 2023, from 6:00 to 20:00. Second, the ENVI-met model was validated using RMSE analysis. Third, 18 scenarios were created to test heat-mitigation strategies, including changes to surface materials (grass, water, concrete) and 3D vegetation (Table 1). Fourth, thermal comfort was assessed using Physiological Equivalent Temperature (PET) and MRT. Lastly, ANOVA and Tukey's HSD tests identified key courtyard features influencing thermal comfort, informing actionable design recommendations.

3 Results

3.1 Model Validation

The validation results from June 20, 2023, comparing ENVI-met (modeled) and MaRTy (observed) data for the Herberger Institute reveal varying accuracies. MRT shows strong alignment (R^2 = 0.95, RMSE = 6.2, MBE = -4.1), and Ta also performs strong alignment with observed values, showcasing high accuracy (R^2 = 0.90, d = 0.98) and minimal error (RMSE = 1.2, MBE = 0.7). RH has moderate accuracy with a weaker correlation (R^2 = 0.49) but acceptable accuracy (d = 0.74). The model tends to overestimate RH (MBE = 7.9). WS shows Limited reliability due to low correlation (R^2 = 0.05) and significant error despite a small RMSE (0.4). MBE suggests a slight underestimation (-0.3), while Tsurface performs strongly (R^2 = 0.88, RMSE = 4.7, MBE = 1.4). The ENVI-met model effectively predicts MRT, Ta, and Tsurface but requires further calibration for RH and WS.

Table 2: Validation results for June 20, 2023, modeled/ENVI-met vs observed/MaRTy for the Herberger Institute

	Herberger Institute for Design and the Arts Hall			
Variables	RMSE	R^2	d	MBE
MRT	6.2	0.95	0.95	-4.1
Ta	1.2	0.90	0.98	0.7
RH	8.8	0.49	0.74	7.9
WS	0.4	0.05	0.61	-0.3
Tsurface	4.7	0.88	0.91	1.4

3.2 Impact of Surface Materials and Vegetation on MRT

The effect of surface materials on MRT was assessed across various courtyard scenarios, revealing that grass surfaces, either alone or with water, were most effective at reducing MRT, while concrete surfaces were less effective at minimizing heat retention. The open grass scenario (S1) achieved the most significant MRT reductions, especially in the morning and evening, with MRT values ranging from 14.7°C at 06:00 h to 58.5-61.2°C at 14:00 h. Adding water (S2) provided only marginal improvements. Concrete scenarios resulted in higher MRT values. Combining concrete with grass (S5) or grass with water (S6) provided moderate cooling but remained less effective than grass alone. The full concrete scenario (S3) had the poorest thermal performance, with MRT peaking between 59.2-63.5°C. In conclusion, grass surfaces were the most effective at reducing MRT, especially when combined

with water. Concrete, even in combination with grass or water, offered minimal heat mitigation. Vegetation placement also influenced MRT. Central vegetation (S7, S9, S13) provided more significant MRT reductions compared to border vegetation (S13-S15), particularly in the early morning, with reductions of 1-3°C. Grass was more effective than concrete, but some water features caused higher midday MRT. Central vegetation was more effective at reducing MRT throughout the day, while border vegetation maintained lower MRT in the evening (Table 3).

Table 3: MRT of the courtyard model comparing different heat mitigation strategies

1st scenario group: Effects of materials															
Scenarios	**6:00**	**7:00**	**8:00**	**9:00**	**10:00**	**11:00**	**12:00**	**13:00**	**14:00**	**15:00**	**16:00**	**17:00**	**18:00**	**19:00**	**20:00**
S0_Existing Situation	18.57	24.08	30.70	36.71	40.75	43.33	46.17	49.99	52.33	51.21	48.03	40.77	36.76	31.23	28.73
S1_FullGrass_No 3D plant	14.74	27.81	41.48	49.38	52.21	53.19	54.06	57.44	61.23	62.44	61.01	52.21	40.28	32.42	26.67
S2_Grass_Water_No 3D plant	14.70	27.77	41.47	49.38	52.21	53.19	54.06	57.44	61.21	62.40	60.96	52.16	40.22	32.34	26.62
S3_FullConcrete_No 3D plant	14.48	27.67	41.58	49.77	52.80	53.95	54.94	58.39	62.09	63.13	61.51	52.56	40.49	32.47	26.67
S4_Concrete_Water_No 3D plant	14.48	27.67	41.55	49.69	52.68	53.80	54.76	58.20	61.90	62.94	61.36	52.43	40.37	32.37	26.59
S5_Concrete_Grass_No 3D plant	14.52	27.70	41.56	49.69	52.68	53.80	54.77	58.21	61.94	63.01	61.42	52.50	40.45	32.46	26.67
S6_Concrete_Water_Grass_No 3D plant	14.57	27.74	41.61	49.74	52.69	53.80	54.76	58.19	61.87	62.91	61.34	52.40	40.33	32.35	26.62
2nd scenario group: Effects of 3D plants with a centered location															
S7_FullGrass_Center_3D plant	17.87	22.10	26.76	33.83	39.44	43.70	47.77	51.52	53.42	52.96	50.32	44.13	39.06	34.30	29.68
S8_Grass_Water_Center_3D plant	14.70	27.77	27.98	35.23	45.58	50.29	54.06	57.44	61.21	62.40	62.40	52.16	40.22	32.34	26.62
S9_FullConcrete_Center_3D plant	17.64	21.98	26.83	34.16	39.93	44.35	48.55	52.40	54.31	53.75	50.88	44.51	39.27	34.35	29.62
S10_Concrete_Water_Center_3D plant	17.63	21.97	26.82	34.14	39.91	44.32	48.51	52.35	54.26	53.66	50.82	44.44	39.19	34.28	29.46
S11_Concrete_Grass_Center_3D plant	17.68	22.00	26.82	39.84	44.23	48.41	52.24	54.14	53.59	50.78	44.44	39.23	34.35	29.62	25.60
S12_Concrete_Water_Grass_Center_3D plant	17.71	22.01	26.80	34.03	39.73	44.10	48.24	52.05	53.95	53.43	50.63	44.33	39.14	34.28	29.53
2nd scenario group: Effects of 3D plants with bordered location															
S13_FullGrass_Border_3D plant	18.37	22.60	27.70	34.75	39.57	43.19	47.58	51.34	52.75	52.00	49.50	44.08	39.29	34.69	30.14
S14_Grass_Water_Border_3D plant	18.33	22.57	27.70	34.78	39.60	43.21	47.60	51.36	52.76	51.98	49.50	44.06	39.25	34.62	30.06
S15_FullConcrete_Border_3D plant	18.13	22.47	27.77	35.08	40.07	43.85	48.35	52.20	53.61	52.75	50.04	44.43	39.48	34.72	30.08
S16_Concrete_Water_Border_3D plant	18.13	22.47	27.76	35.05	40.00	43.75	48.20	52.03	53.44	52.59	49.92	44.33	39.38	34.63	29.96
S17_Concrete_Grass_Border_3D plant	18.18	22.50	27.76	35.02	39.97	43.72	48.19	52.02	53.45	52.62	49.94	44.36	39.44	34.71	30.08
S18_Concrete_Water_Grass_Border_3D plant	17.92	22.33	27.96	36.23	41.11	44.65	48.58	52.32	54.12	53.32	50.48	44.29	39.19	34.45	29.73

3.3 ANOVA for PET

We examined the influence of courtyard design features on thermal comfort during peak PET levels (14:00). To assess statistically significant differences in PET across various scenarios, an ANOVA Tukey HSD test was performed (Figure 2). A comparison of PET levels at the Herberger Institute for Design and the Arts Hall demonstrated notable variations in thermal comfort, shaped by factors such as surface materials and the strategic placement of 3D vegetation. The first group of scenarios showed no substantial impact on PET values, whereas the second group revealed statistically significant differences based on vegetation placement-centralized or bordered. Vegetation reduced PET by 1–2°C, with statistically significant results.

The effect of surface materials (S1–S6) on PET showed considerable variability. At the Herberger Institute, changes in surface materials moderately increased PET, with scenario S5 resulting in the highest rise of +4.9°C compared to the base case. Scenarios involving 3D vegetation (S7–S18) consistently reduced PET across all locations, with centralized vegetation (S7–S12) generally achieving greater reductions than bordered vegetation (S13–S18). Specifically, centralized vegetation at the Herberger Institute reduced PET by up to 4.5°C, while bordered vegetation achieved a reduction of 1.7°C.

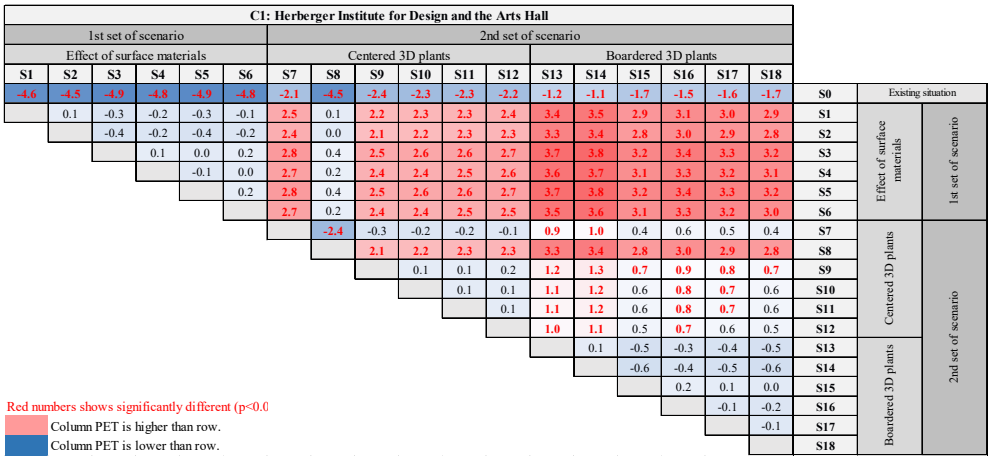


Fig. 2: Tukey-HSD comparison matrix for all scenarios, where the blue and red grids show the differences in the mean PET at 14:00 h

4 Conclusion and Outlook

This study highlights the critical role of integrating green spaces and shading structures to improve thermal comfort in courtyard settings. Surface materials – such as grass, concrete, and water – significantly influenced Mean Radiant Temperature (MRT), though shading structures had a stronger impact. Among surface materials, grass proved the most effective, reducing MRT by 1–2°C compared to concrete. Scenarios combining grass and water demonstrated marginal improvements, while water features alone offered limited enhancements to thermal comfort. Vegetation, particularly trees, reduced MRT by 4–5°C and Physiological Equivalent Temperature (PET) by 1–2°C across various scenarios. However, the effectiveness of vegetation varied depending on its placement and the surface materials used.

Central vegetation was more beneficial during cooler hours but contributed to heat retention during peak midday temperatures. In contrast, perimeter vegetation provided consistent shading without obstructing airflow, making it a more reliable strategy for long-term MRT reduction. Scenarios combining grass, water, and perimeter vegetation consistently delivered the best thermal performance. Conversely, dense central vegetation paired with large concrete surfaces heightened MRT during midday, underscoring the importance of strategic vegetation placement and material selection.

These findings align with previous research emphasizing the need for integrated design strategies that leverage vegetation and surface materials to optimize thermal comfort (BERKOVIC et al. 2012, GHAFFARIANHOSEINI et al. 2015, LIU et al. 2024, LOPEZ-CABEZA 2022, MARCHI et al. 2023). While vegetation consistently demonstrated effectiveness in reducing MRT and PET, the performance of other strategies, such as high-albedo materials and water features, was more context-specific. This study unde

rscores the necessity of tailoring design approaches to the unique microclimatic and spatial characteristics of courtyards to achieve optimal thermal comfort. As global temperatures rise and urban heat island effects intensify, there is an urgent need for urban planners, architects,

and landscape designers to create outdoor spaces that enhance thermal comfort while minimizing energy use and environmental impact. The insights from this study provide valuable guidance for designing courtyards in Phoenix and other hot-arid cities worldwide, where addressing extreme heat is increasingly critical for public health and well-being.

References

AKBARI, H., CHERATI, S. M., MONAZAM, N. H. & NOGUCHI, M. (2021), Effect of courtyards' geometrical parameters on climate adaptability and shading performance in hot-arid climate of Yazd (Iran). Sustainable Energy Technology and Assessments, 48, 101594. https://doi.org/https://doi.org/10.1016/j.seta.2021.101594.

AZMET (2023), Arizona Climate Data. https://ag.arizona.edu/azmet/az-data.htm (09.08.2023).

BERKOVIC, S., YEZIORO, A. & BITAN, A. (2012), Study of thermal comfort in courtyards in a hot arid climate. Solar Energy, 86, 1173-1186. https://doi.org/10.1016/j.solener.2012.01.010.

CILEK, A., UNAL, M. & MIDDEL, A. (2024), The effects of 2-D and 3-D urban landscape metrics on mean radiant temperature in hot-arid Phoenix and Tempe, Arizona, USA. Sustainable Cities and Society, 101, 105116.

CRANK, P. J., MIDDEL, A., WAGNER, M., HOOTS, D., SMITH, M. & BRAZEL, A. (2020), Validation of seasonal mean radiant temperature simulations in hot arid urban climates. Science of the Total Environment, 749, 141392. https://doi.org/10.1016/j.scitotenv.2020.141392.

DIZ-MELLADO, E., LOPEZ-CABEZA, V. P., RIVERA-GOMEZ, C., GALAN-MARIN, C., ROJAS-FERNANDEZ, J. & NIKOLOPOULOU, M. (2021), Extending the adaptive thermal comfort models for courtyards. Building and Environment, 203, 108094. https://doi.org/10.1016/j.buildenv.2021.108094.

GELETIČ, J., LEHNERT, M., SAVIĆ, S. & MILOŠEVIĆ, D. (2019), Inter-/intra-zonal seasonal variability of the surface urban heat island based on local climate zones in three central European cities. Building and Environment, 156 (December 2018), 21-32. https://doi.org/10.1016/j.buildenv.2019.04.011.

GHAFFARIANHOSEINI, A., BERARDI, U. & GHAFFARIANHOSEINI, A. (2015), Thermal performance characteristics of unshaded courtyards in hot and humid climates. Building and Environment, 87, 154-168. https://doi.org/10.1016/j.buildenv.2015.02.001.

HU, J., YANG, Y., PAN, X., ZHU, Q., ZHAN, W., WANG, Y., MA, W. & SU, W. (2019), Analysis of the Spatial and Temporal Variations of Land Surface Temperature Based on Local Climate Zones: A Case Study in Nanjing, China. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 12 (11), 4213-4223. https://doi.org/10.1109/JSTARS.2019.2926502.

LIU, S., MIDDEL, A., FANG, X. & WU, R. (2024), Urban Climate ENVI-met model performance evaluation for courtyard simulations in hot-humid climates. Urban Clim. 2024, 55, 101909.

LOPEZ-CABEZA, V. P., ALZATE-GAVIRIA, S., DIZ-MELLADO, E., RIVERA-GOMEZ, C. & GALAN-MARIN, C, (2022), Albedo influence on the microclimate and thermal comfort of courtyards under Mediterranean hot summer climate conditions. Sustainable Cities and Society, 81, 103872. https://doi.org/10.1016/j.scs.2022.103872.

MARCHI, L., GASPARI, J. & FABBRI, K. (2023), Outdoor Microclimate in Courtyard Buildings: Impact of Building Perimeter Configuration and Tree Density. Buildings, 13, 2687. https://doi.org/https://doi.org/10.3390/buildings13112687.

MIDDEL, A., HÄB, K., BRAZEL, A. J., MARTIN, C. & GUHATHAKURTA, S. (2014), Impact of urban form and design on mid-afternoon microclimate in Phoenix Local Climate Zones. Landscape and Urban Planning, 122, 16-28. https://doi.org/10.1016/j.landurbplan.2013.11.004.

MIDDEL, A. & KRAYENHOFF, E. S. (2019), Micrometeorological determinants of pedestrian thermal exposure during record-breaking heat in Tempe, Arizona: Introducing the MaRTy observational platform. The Science of the total environment, 687, 137-151. https://doi.org/10.1016/j.scitotenv.2019.06.085.

MUSHTAHA, E., SHAREEF, S., ALSYOUF, I., MORI, T., KAYED, A., ABDELRAHIM, M. & ALBANNAY, S. (2021), A study of the impact of major Urban Heat Island factors in a hot climate courtyard : The case of the University of Sharjah, UAE. Sustainable Cities and Society, 69, 102844. https://doi.org/10.1016/j.scs.2021.102844.

SALUR, H. (2016), Avlulu Yapılarda Termal Konfor Analizi: Kayseri Köşk Medrese Örneği. Erciyes Üniversitesi, Fen Bilimleri Enstitüsü, Mimarlık Anabilim Dalı, Doktora Tezi, Kayseri, 106.

SOFLAEI, F., SHOKOUHIAN, M., ABRAVESHDAR, H. & ALIPOUR, A. (2017), The impact of courtyard design variants on shading performance in hot- arid climates of Iran. Energy and Buildings, 143, 71-83. https://doi.org/http://dx.doi.org/10.1016/j.enbuild.2017.03.027.

SOFLAEI, F., SHOKOUHIAN, M., TABADKANI, A., MOSLEHI, H. & BERARDI, U. (2020), A simulation-based model for courtyard housing design based on adaptive thermal comfort. Journal of Building Engineering, 31, 101335. https://doi.org/10.1016/j.jobe.2020.101335.

SÖZEN, İ. & ORAL, G. K. (2019), Outdoor thermal comfort in urban canyon and courtyard in hot arid climate: A parametric study based on the vernacular settlement of Mardin. Sustainable Cities and Society, 48, 101398. https://doi.org/https://doi.org/10.1016/j.scs.2018.12.026.

TALEGHANI, M., TENPIERIK, M., VAN DEN DOBBELSTEEN, A. & SAILOR, D. J. (2014), Heat in courtyards : A validated and calibrated parametric study of heat mitigation strategies for urban courtyards in the Netherlands. Solar Energy, 103, 108-124. https://doi.org/http://dx.doi.org/10.1016/j.solener.2014.01.033.

UNAL CILEK, M. & CILEK, A. (2021), Analyses of land surface temperature (LST) variability among local climate zones (LCZs) comparing Landsat-8 and ENVI-met model data. Sustainable Cities and Society, 69, 102877. https://doi.org/10.1016/j.scs.2021.102877.

UNAL, M. & USLU, C. (2022), Modeling the relationship between the geometric characteristics of urban green spaces and thermal comfort: The case of Adana city. Sustainable Cities and Society, 79, 103748. https://doi.org/10.1016/j.scs.2022.103748.

WU, R., FANG, X., LIU, S. & MIDDEL, A. (2023), A fast and accurate mean radiant temperature model for courtyards: Evidence from the Keyuan Garden in central Guangdong, China. Building and Environment, 229, 109916. https://doi.org/10.1016/j.buildenv.2022.109916.